

Signature Detection and Matching for Document Image Retrieval

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Abstract—As one of the most pervasive methods of individual identification and document authentication, signatures present convincing evidence and provide an important form of indexing for effective document image processing and retrieval in a broad range of applications. However, detection and segmentation of free-form objects such as signatures from clustered background is currently an open document analysis problem. In this paper, we focus on two fundamental problems in signature-based document image retrieval. First, we propose a novel multiscale approach to jointly detecting and segmenting signatures from document images. Rather than focusing on local features that typically have large variations, our approach captures the structural saliency using a signature production model and computes the dynamic curvature of 2D contour fragments over multiple scales. This detection framework is general and computationally tractable. Second, we treat the problem of signature retrieval in the unconstrained setting of translation, scale, and rotation invariant nonrigid shape matching. We propose two novel measures of shape dissimilarity based on anisotropic scaling and registration residual error and present a supervised learning framework for combining complementary shape information from different dissimilarity metrics using LDA. We quantitatively study state-of-the-art shape representations, shape matching algorithms, measures of dissimilarity, and the use of multiple instances as query in document image retrieval. We further demonstrate our matching techniques in offline signature verification. Extensive experiments using large real-world collections of English and Arabic machine-printed and handwritten documents demonstrate the excellent performance of our approaches.

Index Terms—Document image analysis and retrieval, signature detection and segmentation, signature matching, structural saliency, deformable shape, measure of shape dissimilarity.

1 INTRODUCTION

SEARCHING for relevant documents from large complex document image repositories is a central problem in document image analysis and retrieval. One approach is to recognize text in the image using an optical character recognition (OCR) system and apply text indexing and query. This solution is primarily restricted to machine-printed text content because the state-of-the-art handwriting recognition is error prone and limited to applications with a small vocabulary, such as postal address recognition and bank check reading [1]. In broader, unconstrained domains, including searching of historic manuscripts [2] and the processing of languages where character recognition is difficult [3], image retrieval has demonstrated much better results.

As unique and evidentiary entities in a wide range of business and forensic applications, signatures provide an important form of indexing that enables effective exploration of large heterogeneous document image collections. Given an abundance of documents, searching for a specific signature is a highly effective way of retrieving documents authorized or authored by an individual [4]. In this context, handwriting recognition is suboptimal because of its prohibitively low recognition rate and the fact that the character sequence of a signature is often unrelated to the signature itself. More importantly, as a number of studies [5], [6] demonstrated, signatures are highly stylistic in nature and are best described by their graphic style.

Detecting, segmenting, and matching deformable objects such as signatures are important and challenging problems in computer vision. In the following sections, we motivate detection, segmentation, and matching in the context of signature-based document image retrieval and present an overview of our approach.

1.1 Signature Detection and Segmentation

Detecting free-form objects is challenging in a number of aspects. First, detection needs to be robust in the presence of large intraclass variations and cluttered backgrounds. Second, the contours of these complex objects are fragmented 2D signals in real offline images for which reliably recovering the order of points in handwriting is difficult in general [7]. Moreover, recognition and retrieval techniques require well-segmented objects to minimize the effects of outliers during matching. Detecting signatures from documents is one such representative example that further

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involves diverse layout structures and complex mixture of machine-printed text, handwriting, diagrams, and other elements [8]. Signature detection and segmentation is still an open research problem [1].

Prior research on offline signatures has almost exclusively focused on signature verification and identification [5], [6], [9], [10], [11], [12], [13] in the context of biometrics to perform authentication. For signature verification, the problem is to decide whether a sample signature is genuine or a forgery by comparing it with stored reference signatures. Signature identification is essentially a writer identification problem, whose objective is to find the signer of a test sample given a database of signature exemplars from different signers. Most studies published till date assume that good detection and segmentation are available.

Solving the problem of signature detection and segmentation is pivotal for signature-based document indexing and retrieval. Equally important is that a solution will also benefit offline signature verification and identification in a range of domains. In addition, the ability to robustly detect signatures and extract them intact from volumes of documents is highly desirable in many business and government applications.

We propose a new multiscale approach to jointly detecting and extracting signatures from document images. Rather than focusing on local features, we treat signatures as symbols that exhibit characteristic structural saliency in our multiscale detection framework. We employ a novel saliency measure based on a signature production model, which captures the dynamic curvature in a signature without recovering its temporal information. The signature detection approach can also be applied to online handwritten notes, where the trajectories of the pen are readily available.

Our detection framework is general and has the advantage that it does not embed explicit assumptions on local features of signatures such as the granulometric size distributions [9] or stroke-level features [5]. Therefore, it is robust against many forms of variations that are challenges in shape-based object detection problems and generally applicable in spite of differences across languages.

1.2 Signature Matching for Document Image Retrieval

Detection and segmentation produce a set of 2D contour fragments for each detected signature. Given a few available query signature instances and a large database of detected signatures, the problem of signature matching is to find the most similar signature samples from the database. By constructing the list of best matching signatures, we effectively retrieve the set of documents authorized or authored by the same person.

We treat a signature as a nonrigid shape and represent it by a discrete set of 2D points sampled from the internal or external contours on the object. Two-dimensional point feature offers several competitive advantages compared to other compact geometrical entities used in shape representation because it relaxes the strong assumption that the topology and the temporal order need to be preserved under structural variations or clustered background. For instance, two strokes in one signature sample may touch each other but remain well separated in another. These

structural changes, as well as outliers and noise, are generally challenging for shock-graph-based approaches [14], [15], which explicitly make use of the connection between points. In some earlier studies [16], [17], [18], [19], a shape is represented as an ordered sequence of points. This 1D representation is well suited for signatures collected online using a PDA or Table PC. For unconstrained offline handwriting in general, however, it is difficult to recover their temporal information from real images due to the large structural variations [7]. Represented by a 2D point distribution, a shape is more robust under structural variations while carrying general shape information. As shown in Fig. 2, the shape of a signature is well captured by a finite set $\mathcal{P} = \{P_1, \dots, P_n\}$, $P_i \in \mathbb{R}^2$, of n points, which are sampled from edge pixels computed by an edge detector.¹

We use two state-of-the-art nonrigid shape matching algorithms for signature matching. The first method is based on the representation of shape contexts, introduced by Belongie et al. [20]. In this approach, a spatial histogram defined as *shape context* is computed for each point, which describes the distribution of the relative positions of all remaining points. Prior to matching, the correspondences between points are solved first through weighted bipartite graph matching. Our second method uses the nonrigid shape matching algorithm proposed by Zheng and Doermann [21], which formulates shape matching as an optimization problem that preserves local neighborhood structure. This approach has an intuitive graph matching interpretation, where each point represents a vertex and two vertices are considered connected in the graph if they are neighbors. The problem of finding the optimal match between shapes is thus equivalent to maximizing the number of matched edges between their corresponding graphs under a one-to-one matching constraint.² Computationally, the authors of [21] employ an iterative framework for estimating the correspondences and the transformation. In each iteration, graph matching is initialized using the shape context distance and subsequently updated through relaxation labeling [22] for more globally consistent results.

Treating an input pattern as a generic 2D point distribution broadens the space of dissimilarity metrics and enables effective shape discrimination using the correspondences and the underlying transformations. We propose two novel shape dissimilarity metrics that quantitatively measure anisotropic scaling and registration residual error, and present a supervised training framework for effectively combining complementary shape information from different dissimilarity measures by linear discriminant analysis (LDA).

The contributions of this paper are twofold:

1. We propose a novel multiscale approach to jointly detecting and segmenting signatures from documents with diverse layouts and complex backgrounds. We treat a signature generally as an unknown grouping of 2D contour fragments and solve for the two unknowns—identification of the

1. We randomly select these n sample points from the contours via a rejection sampling method that spreads the points over the entire shape.

2. To robustly handle outliers, multiple points are allowed to match to the dummy point added to each point set.

most salient structure and its grouping, using a signature production model that captures the dynamic curvature of 2D contour fragments without recovering the temporal information.

2. We treat signature recognition and retrieval in the unconstrained setting of nonrigid shape matching and propose two new measures of shape dissimilarity that correspond to a) the amount of anisotropic scaling and b) the registration residual error. We demonstrate robust signature-based document image retrieval and comprehensively evaluate different shape representations, shape matching algorithms, measures of dissimilarity, and the use of multiple signature instances in overall retrieval accuracy.

The structure of this paper is as follows: The next section reviews related work. In Section 3, we present our multi-scale structural saliency approach to signature detection and segmentation in detail. Section 4 introduces a saliency measure for capturing the dynamic curvature using a signature production model. In Section 5, we describe our signature matching approach and present methods to combine different measures of shape dissimilarity and multiple query instances for effective retrieval with limited supervised training. We discuss experimental results on real English and Arabic document data sets in Section 6 and conclude our paper in Section 7.

2 RELATED WORK

2.1 Structural Saliency and Contour Grouping

Detecting salient structures from images is an important task in many segmentation and recognition problems. The *saliency network* proposed by Sha'ashua and Ullman [23] is a well-known approach to extracting salient curves by jointly solving the two aspects of this problem iteratively, i.e., identifying salient structures and grouping contours. The saliency function defined in [23] monotonically increases with the length and monotonically decreases with the total squared curvature of the evaluated curve. To reduce the exponential search space, the saliency network assumed that an optimal solution has a recurrent structure, which they called *extensibility*, so that searching by dynamic programming in the exponential space of possible curves takes polynomial time. However, greedily reducing the solution space by such a recurrent formulation involves hard decision at each step, and theoretically, a single mistake can result in the convergence to a wrong solution.

Alter and Basri [24] presented a comprehensive analysis of the saliency network and derived its $O(k^2 N^2)$ time complexity, where N is the total number of pixels and k is the number of neighboring elements considered in forming a locally connected network. They demonstrated that the salient network has a few serious problems due to the extensibility assumption and the convergence rates vary significantly depending on the object structure. These limitations are difficult to overcome without fundamentally changing the computation.

There is a body of literature on contour grouping and contour-based learning in computer vision. Here, we point out some of the work more related to ours, which includes

Parent and Zucker's [25] work using relaxation methods and Guy and Medioni's [26] work using voting patterns. Elder and Zucker [27] developed a method for finding closed contours using chains of tangent vectors. Williams and Jacobs [28] and Williams and Thornber [29] discussed contour closure using stochastic completion fields. Shotton et al. [30] demonstrated a learning-based method for object detection using local contour-based features extracted from a single image scale. Such an approach, however, is inherently restricted to closed contour shapes for which the ordering of points is explicit.

2.2 Shape Matching

Rigid shape matching has been approached in a number of ways with intent to obtain a discriminative global shape description. Approaches using silhouette features include Fourier descriptors [31], [32], geometric hashing [33], dynamic programming [17], [34], and skeletons derived using Blum's medial axis transform [35]. Although silhouettes are simple and efficient to compare, they are limited as shape descriptors because they ignore internal contours and are difficult to extract from real images [36]. Other approaches, such as chamfer matching [37] and the Hausdorff distance [38], treat the shape as a discrete set of points extracted using an edge detector. Unlike approaches that compute correspondences, these methods do not enforce pairing of points between the two sets being compared. While they work well under a selected subset of rigid transformations, they cannot generally be extended to handle nonrigid transformations. The reader may refer to [39], [40] for a general survey on classic rigid shape matching techniques.

Matching for nonrigid shapes needs to consider unknown transformations that are both linear (e.g., translation, rotation, scaling, and shear) and nonlinear. One comprehensive framework for shape matching in this general setting is to iteratively estimate the correspondence and the transformation. The iterative closest point (ICP) algorithm introduced by Besl and McKay [41] and its extensions [42], [43], [44] provide a simple heuristic approach. Assuming that two shapes are roughly aligned, the nearest neighbor in the other shape is assigned as the estimated correspondence at each step. This estimate of the correspondence is then used to refine the estimated affine or piecewise-affine mapping, and vice versa. While ICP is fast and guaranteed to converge to a local minimum, its performance degenerates quickly when large nonrigid deformation or a significant amount of outliers are involved [45]. Chui and Rangarajan [46] developed an iterative optimization algorithm to jointly estimate the correspondences and the transformation, using thin plate splines as a generic parameterization of a nonrigid transformation. Joint estimation of correspondences and transformation leads to a highly nonconvex optimization problem, which is solved using the softassign and deterministic annealing.

2.3 Document Image Retrieval

Rath et al. [47] demonstrated retrieval of handwritten historical manuscripts by using images of handwritten words to query unlabeled document images. The system compares word images based on Fourier descriptors

computed from a collection of shape features, including the projection profile and the contours extracted from the segmented word. Mean average precision of 63 percent was reported for image retrieval when tested using 20 images by optimizing two-word queries.

Srihari et al. [48] developed a signature matching and retrieval approach by computing correlation of gradient, structural, and concavity features extracted from fixed-size image patches. It achieved 76.3 percent precision using a collection of 447 manually cropped signature images from the Tobacco-800 database [49], [50]. The approach is not translation, scale, or rotation invariant.

3 MULTISCALE STRUCTURAL SALIENCY

3.1 Theoretical Framework

We consider the identification of salient structure and the grouping of its structural components separately. There are clear motivations for decoupling these two unknowns as opposed to solving them jointly. First, we have a broader set of functions that can be used as measures of saliency. For object detection, using saliency measures that fit high-level knowledge of the object gives globally more consistent results than jointly optimizing a fixed set of low-level vision constraints. Once the salient structures are identified, the problem of grouping becomes simpler based on constraints such as proximity and good continuation. Second, we can effectively formulate structural saliency across image scales, as opposed to single-scale approaches such as the saliency network. Multiscale detection is important for nonrigid objects like signatures, whose contours can severely be broken due to the poor ink condition and image degradations. Finally, multiscale saliency computation generates detection hypotheses at the natural scale, where grouping among a set of connected components becomes structurally obvious. These provide a unified framework for object detection and segmentation that produces meaningful representation for object recognition and retrieval.

From a computational point of view, computing saliency using connected components can make the computation tractable and highly parallelizable. Our serial implementation runs in $O(N)$, where N is the total number of edge points. This is significantly faster than the saliency network approach that has $O(k^2 N^2)$ time complexity. We also explore document context to improve detection. The idea is to estimate the length and interline spacing of text lines and use the information to locate the bottom or end of a document, where signatures are more likely to appear. In our evaluation, we show results of signature detection run on whole documents as well as by exploring document context.

3.2 Signature Detection and Segmentation

Signatures exhibit large variations compared to other forms of handwriting because they functionally serve as highly stylish entities for representing individuals [5]. This greatly undercuts the possibility of posing signature detection or recognition as a straightforward classification problem. As shown in Fig. 3, local features, including size, aspect ratio, and spatial density, are not discriminative enough to separate signatures (red pluses) from nonsignature objects (blue dots), on a groundtruthed collection of 1,290 real-world documents.

In this section, we describe a structural saliency approach to signature detection by searching over a range of scales $\mathcal{S} = \sigma_1, \sigma_2, \dots, \sigma_n$. We select the initial scale σ_1 based on the resolution of the input image. We define the multiscale structural saliency for a curve Γ as

$$\Phi(\Gamma) = \max_{\sigma_i \in \mathcal{S}} f(\Phi_{\sigma_i}(\Gamma_{\sigma_i}), \sigma_i), \quad (1)$$

where $f: \mathbb{R}^2 \rightarrow \mathbb{R}^+$ is a function that normalizes the saliency over its scale and Γ_{σ_i} is the corresponding curve computed at the scale σ_i . Using multiple scales for detection relaxes the requirement that the curve Γ be well connected at a specific scale.

Detection at a particular scale σ_i proceeds in three steps. First, we convolve the image with a Gaussian kernel G_{σ_i} , resample it using the Lanczos filter [51] at the factor d_{σ_i} , and compute its edges using the Canny edge detector [52]. This effectively computes a coarse representation of the original image in which small gaps in the curve are bridged by smoothing followed by resampling (see Fig. 1). Second, we form connected components on the edge image at the scale σ_i , and compute the saliency of each component using the measure presented in Section 4, which characterizes its dynamic curvature. We define the saliency of a connected component $\Gamma_{\sigma_i}^k$ as the sum of saliency values computed from all pairs of edges on it. Third, we identify the most salient curves and use a grouping strategy based on proximity and curvilinear constraints to obtain the rest of the signature parts within their neighborhood. Our joint detection and segmentation approach considers identifying the most curvilinear structure and grouping it with neighboring elements in two steps. By locating the most salient signature component, we effectively focus our attention on its neighborhood. Subsequently, a complete signature is segmented out from background by grouping salient neighboring structures.

4 MEASURE OF SALIENCY FOR SIGNATURES

In this section, we consider the problem of recognizing the structural saliency of a 2D offline signature segment using a signature production model. As shown in Fig. 4, among the infinite number of geometric curves that pass two given end points E_1 and E_2 on a signature, very few are realistic. This is because the wrist is highly constrained in the degrees of freedom when producing a signature. Furthermore, a signature segment rarely fits locally to a high-order polynomial, as shown by the dotted curve in Fig. 4.

We propose a signature production model that incorporates two degrees of freedom in the Cartesian coordinates. We assume that the pen moves in a cycloidal fashion with reference to a sequence of shifting baselines when signing a signature. The local baseline changes as the wrist adjusts its position with respect to the document. Within a short curve segment, we assume that the baseline remains unchanged. In addition, the locus of the pen maintains a proportional distance from the local center point (*focus*) to the local baseline (*directrix*), which imposes an additional constraint that limits the group of second-order curves to ellipses. A signature fragment can thus be equivalently viewed as concatenations of small elliptic segments. In a similar spirit, Saint-Marc et al. [53] have used piecewise

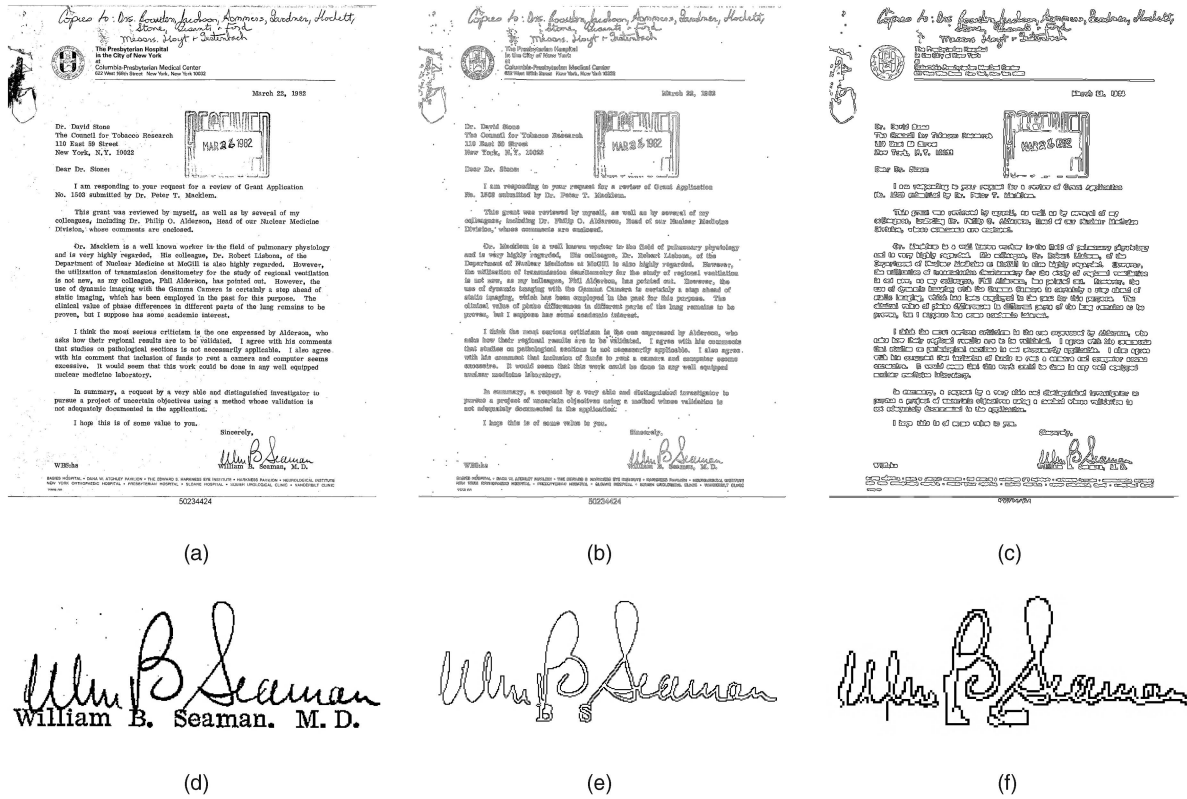


Fig. 1. The prominence of a signature is perceived across scales, whereas background text merges into blobs at coarser scales. (a) A document image example. (b) and (c) are edges computed at the scale of 1.0 and 2.0, respectively. (d) Detected signature region without segmentation. (e) and (f) are detected and segmented signatures by our approach at the two scales.

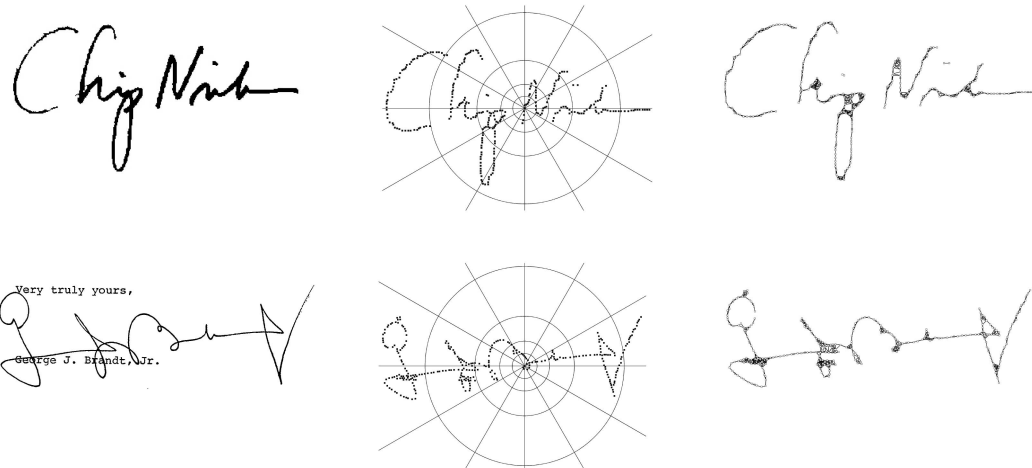


Fig. 2. Shape contexts [20] and local neighborhood graphs [21] constructed from detected and segmented signatures. First column: Examples of signatures. Second column: Shape contexts descriptors constructed at a point which provides a large-scale shape description. Third column: Local neighborhood graphs capture local structures for nonrigid shape matching.

quadratic B-splines to approximate complex-shaped contours for symmetry detection.

Our hypothesis on cycloidal writing is motivated by the oscillation theory of handwriting by Hollerbach [54], who discovered that embedded oscillations coupled along the horizontal and vertical directions produce letter forms that closely resemble handwriting samples. In fact, when the signature baseline is aligned to the x axis, a general form of velocities in the x and y directions is

$$\begin{aligned} v_x &= a \sin(\omega_x t + \phi_x) + c, \\ v_y &= b \sin(\omega_y t + \phi_y), \end{aligned} \quad (2)$$

where a and b are horizontal and vertical velocity amplitudes, ω_x and ω_y are the horizontal and vertical frequencies, ϕ_x and ϕ_y are the horizontal and vertical phases, t is the time variable, and constant c is the average speed of horizontal movement. Without loss of generality, consider the case of $a = b$, $\omega_x = \omega_y$, and $\phi_x - \phi_y = \pi/2$. We can show that, for different values of a and c , the resulting

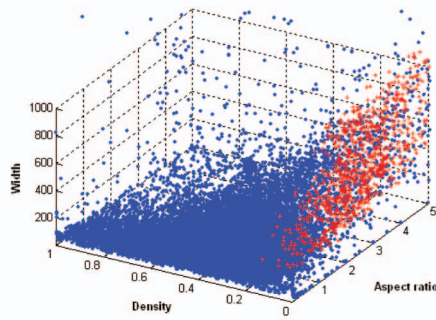


Fig. 3. Nonsignature objects (blue dots) and signatures (red pluses) are nonseparable by local features, including size, aspect ratio, and spatial density, as a significant portion of them overlaps in the feature space.

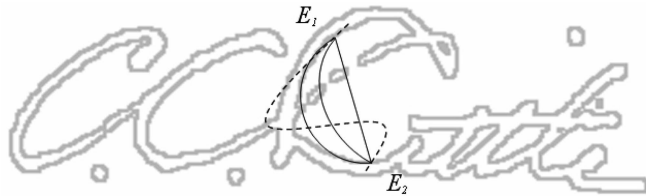


Fig. 4. Among the large number of geometric curves passing through the two end points E_1 and E_2 on a signature, a few are realistic (solid curves). An example of unrealistic curve is shown in the dotted line.

curves are curtate cycloid, cycloid, and prolate cycloid, as shown in Fig. 5, respectively.

We model piecewise segments of a signature by a family of second-order curves that satisfy constraints imposed by the signature production. In addition, we use the local gradient directions approximated at the two end points, which can be viewed as soft constraints on the segment of the curve imposed by the global structure of the signature instance. In Cartesian coordinates, the family of a quadratic equation in two variables x and y is always a conic section. Fig. 6 shows how the orientations of the gradients at the two edge points greatly limit the inference on local curve segment to a family of conics, under the second-order signature production model.

We can formalize this intuition geometrically. For a pair of edge points E_1 at (x_1, y_1) and E_2 at (x_2, y_2) , we obtain estimates of their local gradients $N_1(p_1, q_1)$ and $N_2(p_2, q_2)$ through edge detection. For definiteness, we suppose both E_1 and E_2 point into the angular section between the tangent lines, as shown in Fig. 6:

$$p_1(x_2 - x_1) + q_1(y_2 - y_1) > 0 \quad \text{and} \quad (3a)$$

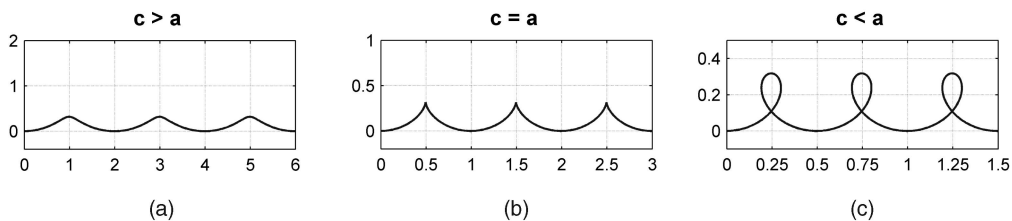


Fig. 5. (a) Curtate cycloid, (b) cycloid, and (c) prolate cycloid curves generated when the speeds of writing in the horizontal baseline and the vertical direction both vary sinusoidally.

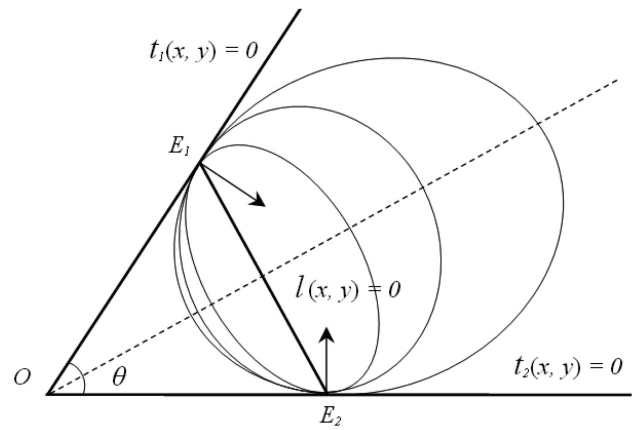


Fig. 6. Conic sections inferred by a pair of edge points.

$$p_2(x_2 - x_1) + q_2(y_2 - y_1) < 0. \quad (3b)$$

The two tangent lines at E_1 and E_2 are normal to their local gradients and are given by

$$t_1(x, y) \equiv p_1(x - x_1) + q_1(y - y_1) = 0 \quad (4)$$

and

$$t_2(x, y) \equiv p_2(x - x_2) + q_2(y - y_2) = 0. \quad (5)$$

The straight line $l(x, y)$ that passes through E_1 and E_2 can be written as

$$l(x, y) \equiv \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0. \quad (6)$$

Note that $t_1(x, y)$, $t_2(x, y)$, and $l(x, y)$ are all first-order linear functions in x and y . The family of second-order curves that is bounded within the angular section between $t_1(x, y)$ and $t_2(x, y)$ can be expressed as

$$C(x, y) \equiv l^2(x, y) - \lambda t_1(x, y)t_2(x, y) = 0. \quad (7)$$

In the canonical form, that is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0, \quad (8)$$

where parameters a, b, c, f, g, h are first-order linear functions in λ and the parameter set (x_1, y_1) , (x_2, y_2) , (p_1, q_1) , (p_2, q_2) .

Given a parameter set (x_1, y_1) , (x_2, y_2) , (p_1, q_1) , (p_2, q_2) , which is equivalent to fixing the set of three straight lines $t_1(x, y)$, $t_2(x, y)$, and $l(x, y)$, we can theoretically analyze how parameter λ for $\lambda \in [0, +\infty)$ affects the curvature of the

bounded quadratic curve defined by (7). When $\lambda = 0$, (7) degenerates into the straight line $l(x, y)$ and the total squared curvature of the bounded segment is strictly zero. When λ monotonically increases from 0 within certain range ($0 < \lambda < \lambda_0$), the curve segment bounded by E_1 and E_2 exhibits more and more curvature. This is because the second-order curves governed by (7) for $\lambda \in [0, \lambda_0]$ are ellipses with monotonically increasing eccentricities. As $\lambda \rightarrow \lambda_0$, the center of the ellipse recedes to infinity so that the ellipse tends to a parabola at $\lambda = \lambda_0$. When $\lambda > \lambda_0$, the conic of (7) becomes a hyperbola. Eventually, as $\lambda \rightarrow +\infty$, the hyperbola degenerates into the two intersected straight lines $t_1(x, y)$ and $t_2(x, y)$. We prove in the Appendix that λ_0 is given by

$$\lambda_0 = \frac{4[p_1(x_2 - x_1) + q_1(y_2 - y_1)]}{(p_1q_2 - p_2q_1)} \times \frac{[p_2(x_1 - x_2) + q_2(y_1 - y_2)]}{(p_1q_2 - p_2q_1)}. \quad (9)$$

Equation (9) provides a theoretical second-order upper bound of the dynamic curvature, given the parameter set $(x_1, y_1), (x_2, y_2), (p_1, q_1), (p_2, q_2)$ that fits the signature production model. We use λ_0 as the saliency value $\Lambda_{\sigma_i}(E_i, E_j)$ for a pair of points at scale σ_i . When $a = |OE_1| = |OE_2|$, it is straightforward to show that the right-hand side of (9) is $4a^2$. This result allows us to normalize saliency over scale, whereas the scale interpretation of most published saliency measures surveyed in [29] is largely unclear. Obviously, our saliency measure is translation and rotation invariant as it only uses local gradient directions.

The saliency of a curve $\Gamma_{\sigma_i}^k$ at scale σ_i is defined as the sum of saliency values computed from all pairs of points on it and is written as

$$\Phi_{\sigma_i}(\Gamma_{\sigma_i}^k) = \sum_{E_i, E_j \in \Gamma_{\sigma_i}^k} \Lambda_{\sigma_i}(E_i, E_j). \quad (10)$$

It intuitively measures the likelihood of elliptic segment fitting given a set of 2D points and the computation does not require knowing the temporal order among points.

The analysis so far applies only to the continuous case. To account for the discretization effect, we impose two additional conditions. First, the absolute values of the two functions on the left-hand side of (3a and 3b) must be strictly large than ϵ . Second, the denominator term in (9) must be strictly large than ϵ . In our experiments, we use $\epsilon = 0.1$. For robustness, we weight the saliency contribution by the gradient magnitude of the weaker edge.

Separating saliency detection from grouping significantly reduces the level of complexity. Let the total number of edge points be N and the average length of connected components be L_c . Saliency computation for each component in (10) requires $O(L_c^2)$ time, on average. Therefore, the overall computation is of order $(N/L_c) \times L_c^2 = NL_c$. Since L_c is effectively upper bounded by prior estimate of signature dimensions and the range of searched scales n is limited, they can be considered as constants. The complexity in saliency computation is linear in N . Gaussian smoothing and connected component analysis both require $O(N)$ time. The total complexity in the signature detection algorithm is therefore $O(N)$.

5 MATCHING AND RETRIEVAL

5.1 Measures of Shape Dissimilarity

Before we introduce two new measures of dissimilarity for general shape matching and retrieval, we first discuss existing shape distance metrics. Each of these dissimilarity measures captures certain shape information for effective discrimination. In the next section, we describe how to effectively combine these individual measures with limited supervised training and present our evaluation framework.

Several measures of shape dissimilarity have demonstrated success in object recognition and retrieval. One is the thin-plate spline bending energy D_{be} and another is the shape context distance D_{sc} . As a conventional tool for interpolating coordinate mappings from $\mathbb{R}^2$ to $\mathbb{R}^2$ based on point constraints, the thin-plate spline (TPS) is commonly used as a generic representation of nonrigid transformation [55]. The TPS interpolant $f(x, y)$ minimizes the bending energy

$$\iint_{\mathbb{R}^2} \left[\left(\frac{\partial^2 f}{\partial x^2} \right)^2 + 2 \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 + \left(\frac{\partial^2 f}{\partial y^2} \right)^2 \right] dx dy \quad (11)$$

over the class of functions that satisfy the given point constraints. Equation (11) imposes smoothness constraints to discourage nonrigidities that are too arbitrary. The bending energy D_{be} [46] measures the amount of *nonlinear* deformation to best warp the shapes into alignment and provides physical interpretation. However, D_{be} only measures the deformation beyond an affine transformation and its functional in (11) is zero if the undergoing transformation is purely affine.

The shape context distance D_{sc} between a template shape $\mathcal{T}$ composed of m points and a deformed shape $\mathcal{D}$ of n points is defined in [20] as

$$D_{sc}(\mathcal{T}, \mathcal{D}) = \frac{1}{m} \sum_{t \in \mathcal{T}} \arg \min_{d \in \mathcal{D}} C(T(t), d) + \frac{1}{n} \sum_{d \in \mathcal{D}} \arg \min_{t \in \mathcal{T}} C(T(t), d), \quad (12)$$

where $T(\cdot)$ denotes the estimated TPS transformation and $C(\cdot, \cdot)$ is the cost function for assigning correspondence between any two points. Given two points, t in shape $\mathcal{T}$ and d in shape $\mathcal{D}$, with associated shape contexts $h_t(k)$ and $h_d(k)$, for $k = 1, 2, \dots, K$, respectively, $C(t, d)$ is defined using the χ^2 statistic as

$$C(t, d) \equiv \frac{1}{2} \sum_{k=1}^K \frac{[h_t(k) - h_d(k)]^2}{h_t(k) + h_d(k)}. \quad (13)$$

We introduce a new measure of dissimilarity D_{as} that characterizes the amount of anisotropic scaling between shapes. Anisotropic scaling is a form of affine transformation that involves change to the relative directional scaling. As illustrated in Fig. 7, the stretching or squeezing of the scaling in the computed affine map captures global mismatch in shape dimensions among all registered points, even in the presence of large intraclass variation.

We compute the amount of anisotropic scaling between two shapes by estimating the ratio of the two scaling factors S_x and S_y in the x and y -directions, respectively. A TPS

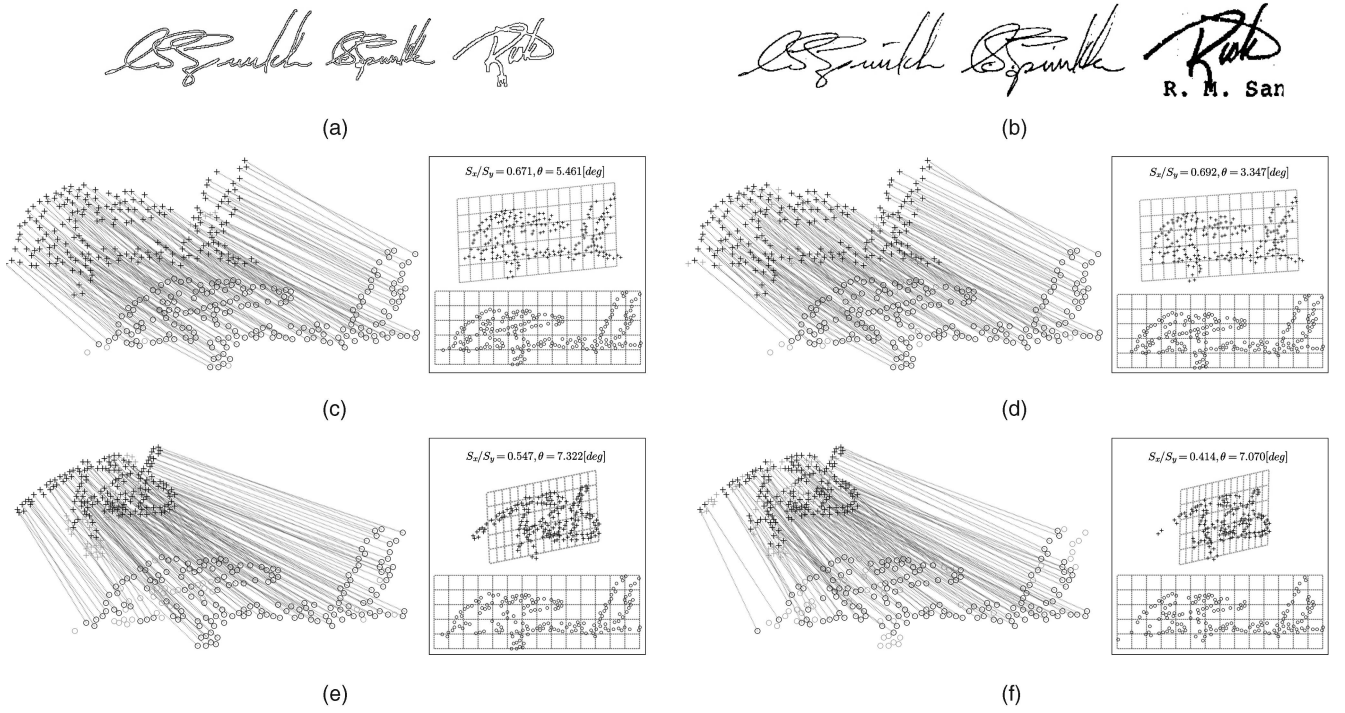


Fig. 7. Anisotropic scaling and registration quality effectively capture shape differences. (a) Signature regions without segmentation. The first two signatures are from the same person, whereas the third one is from a different individual. (b) Detected and segmented signatures by our approach. Second row: Matching results of first two signatures using (c) shape contexts and (d) local neighborhood graph, respectively. Last row: Matching results of the first and third signatures using (e) shape contexts and (f) local neighborhood graph, respectively. Corresponding points identified by shape matching are linked and unmatched points are shown in green. The computed affine maps are shown in figure legends.

transformation can be decomposed into a linear part corresponding to a global affine alignment, together with the superposition of independent, affine-free deformations (or principal warps) of progressively smaller scales [55]. We ignore the nonaffine terms in the TPS interpolant when estimating S_x and S_y . The 2D affine transformation is represented as a 2×2 linear transformation matrix $\mathbf{A}$ and a 2×1 translation vector $\mathbf{T}$

$$\begin{pmatrix} u \\ v \end{pmatrix} = \mathbf{A} \begin{pmatrix} x \\ y \end{pmatrix} + \mathbf{T}, \quad (14)$$

where we can compute S_x and S_y by singular value decomposition on matrix $\mathbf{A}$.

We define D_{as} as

$$D_{as} = \log \frac{\max(S_x, S_y)}{\min(S_x, S_y)}. \quad (15)$$

Note that we have $D_{as} = 0$ when only isotropic scaling is involved (i.e., $S_x = S_y$).

We propose another distance measure D_{re} based on the registration residual errors under the estimated nonrigid transformation. To minimize the effect of outliers, we compute the registration residual error from the subset of points that have been assigned correspondence during matching and ignore points matched to the dummy point nil . Let function $M: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ define the matching between two point sets of size n representing the template shape T and the deformed shape D . Suppose that t_i and $d_{M(i)}$ for $i = 1, 2, \dots, n$ denote pairs of matched points in shape T and shape D , respectively. We define D_{re} as

$$D_{re} = \frac{\sum_{i: M(i) \neq nil} \|T(t_i) - d_{M(i)}\|}{\sum_{i: M(i) \neq nil} 1}, \quad (16)$$

where $T(\cdot)$ denotes the estimated TPS transformation and $\|\cdot\|$ is the euclidean norm.

5.2 Shape Distance

After matching, we compute the overall shape distance for retrieval as the weighted sum of individual distances given by all the measures: shape context distance, TPS bending energy, anisotropic scaling, and registration residual errors:

$$D = w_{sc}D_{sc} + w_{be}D_{be} + w_{as}D_{as} + w_{re}D_{re}. \quad (17)$$

The weights in (17) are optimized by LDA using only a small amount of training data.

The retrieval performance of a single query instance may depend largely on the instance used for the query [56]. In practice, it is often possible to obtain multiple signature samples from the same person. This enables us to use them as an equivalence class to achieve better retrieval performance. When multiple instances $q_1, q_2, \dots, q_k$ from the same class Q are used as queries, we combine their individual distances $D_1, D_2, \dots, D_k$ into one shape distance as

$$D = \min(D_1, D_2, \dots, D_k). \quad (18)$$

5.3 Evaluation Methodology

We use the two most commonly cited measures, average precision and R-precision, to evaluate the performance of each ranked retrieval. Here, we first make precise the intuitions of these evaluation metrics, which emphasize the retrieval ranking differently. Given a ranked list of

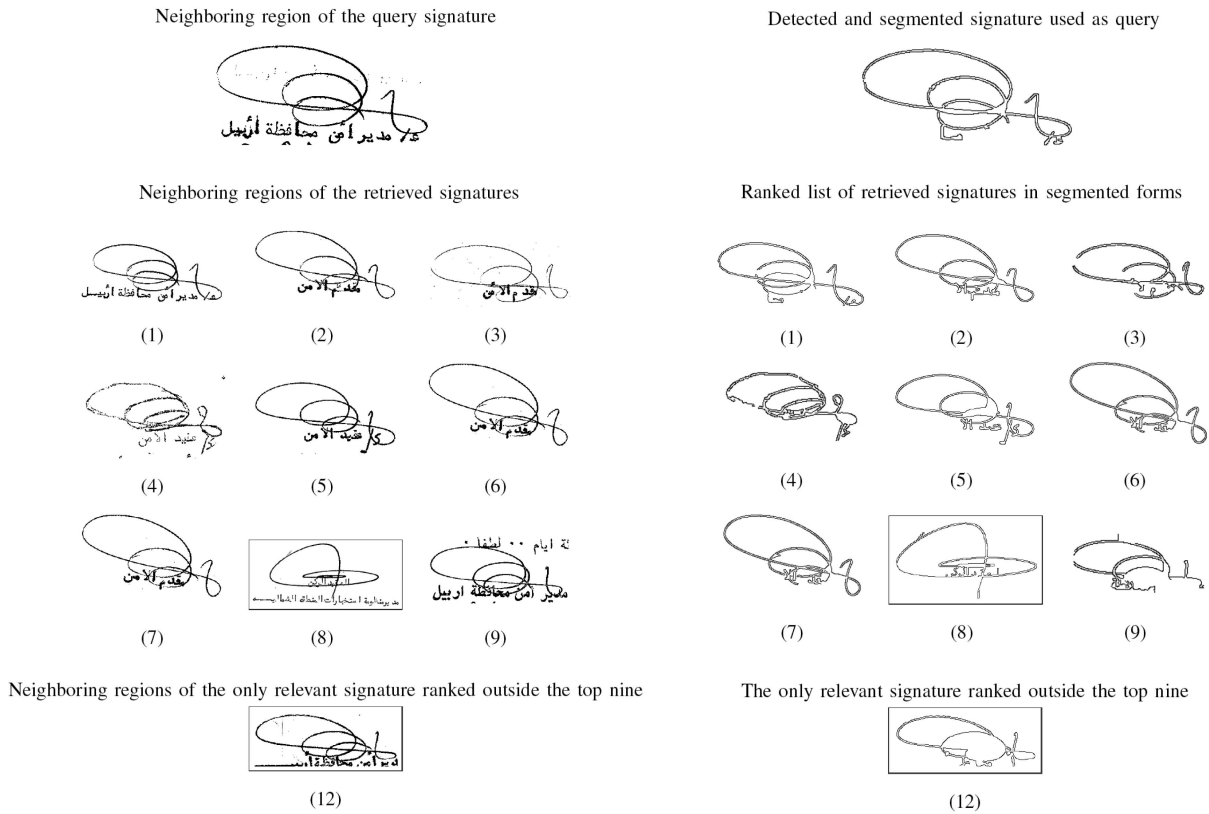


Fig. 8. A signature query example. Among the total of nine relevant signatures, eight appear in the top nine of the returned ranked list, giving average precision of 96.0 percent, and R-precision of 88.9 percent. The irrelevant signature is highlighted by a bounding box. Left: Signature regions in the document. Right: Detected and segmented signatures used in retrieval.

documents returned in response to a query, average precision (AP) is defined as the average of the precisions at all relevant documents. It effectively combines the precision, recall, and relevance ranking, and is often considered as a stable and discriminating measure of the quality of retrieval engines [56] because it rewards retrieval systems that rank relevant documents higher and, at the same time, penalizes those that rank irrelevant ones higher. R-precision (RP) for a query i is the precision at rank $R(i)$, where $R(i)$ is the number of documents relevant to query i . R-precision deemphasizes the exact ranking among the retrieved relevant documents and is more useful when there are large number of relevant documents.

Fig. 8 shows a query example in which eight of the nine total relevant signatures are among the top nine and one relevant signature is ranked 12 in the ranked list. For this query, $AP = (1 + 1 + 1 + 1 + 1 + 1 + 1 + 8/9 + 9/12)/9 = 96.0\%$ and $RP = 8/9 = 88.9\%$.

6 EXPERIMENTS

6.1 Data Sets

To quantitatively evaluate detection and matching in signature-based document image retrieval, we used two large collections of real-world documents—the *Tobacco-800* database [50] and the *University of Maryland Arabic* database [57]. Tobacco-800 is a public subset of the complex document image processing (CDIP) collection [49] and has been used in the TREC 2006 and 2007 evaluations. It was

constructed from 42,000,000 pages of English documents (in 7,000,000 multipage TIFF images) released by tobacco companies and originally hosted at UCSF [58]. Tobacco-800 is a realistic data set for document analysis and retrieval as these documents were collected and scanned using a wide variety of equipments over time. In addition, a significant percentage of Tobacco-800 are multipage business documents, making it a valuable testbed for various content-based image retrieval systems. The Maryland Arabic data set consists of 166,071 Arabic handwritten business documents. Fig. 9 shows some examples from the two data sets.

We tested our system using all the 66 and 21 signature classes in Tobacco-800 and Maryland Arabic data sets among which the number of signatures per person varies in the range from 6 to 11. Table 1 summarizes the important statistics of both evaluation data sets. The overall system performance across all queries is computed quantitatively in mean average precision (MAP) and mean R-precision (MRP), respectively.

6.2 Multiscale Signature Detection

We organize our experiments and evaluation on signature detection as follows. First, we use the detection probability P_D and false-alarm probability P_F as evaluation metrics. P_D and P_F represent the two degrees of freedom in a binary hypothesis test and do not involve a prior probabilities of the hypothesis. To factor in the “quality” of detection, we consider a signature *correctly detected* and *complete* if the

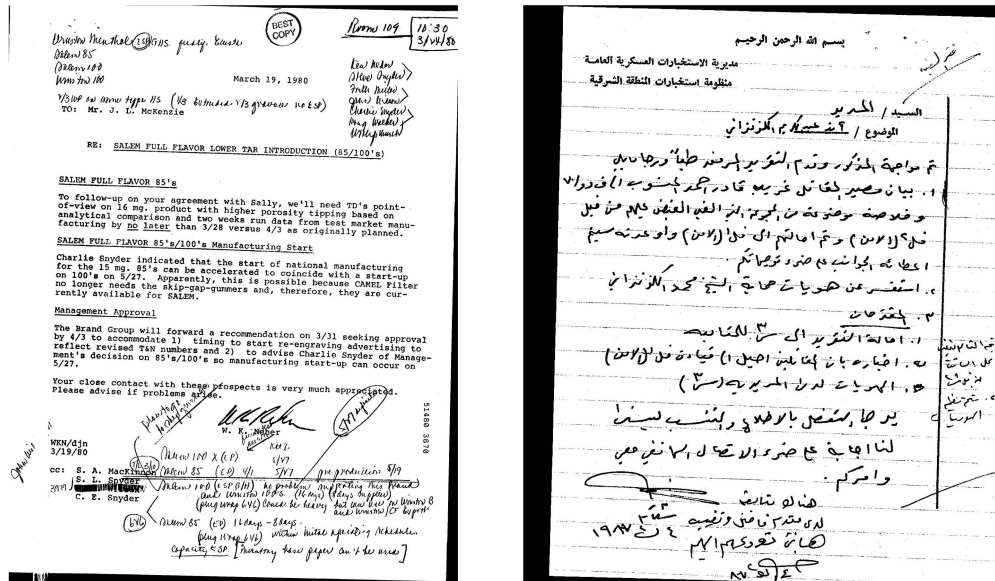


Fig. 9. Examples from (left) the Tobacco-800 database [49], [50] and (right) the Maryland Arabic database [57].

detected region overlaps with more than 75 percent of the labeled signature region. We declare a *false alarm* if the detected region does not overlap with more than 25 percent of any labeled signature. Since the number of signatures varies significantly across documents, in practice, we assume no prior knowledge on the distribution of signatures per document in our experiments.

Fig. 10 shows the receiver operating characteristic (ROC) curves on the Tobacco-800 and Maryland Arabic data sets. A Fisher classifier using size, aspect ratio, and spatial density features serves as a baseline for comparison, with all other procedures remaining the same in the experiment. We use two scale levels in multiscale detection experiments. Parameters involved in obtaining the ROC curves, including the detection threshold in saliency and estimates of signature dimensions, are tuned on 10 documents. We use the following approach to compute each operating point on an ROC curve. We sort the list of detected candidates from the entire test set by their saliencies. To plot a new point, we move down the ranked list by one and look at the portion of the ranked list from top to the current rank, thus equivalent to gradually lowering the decision threshold. The entire set of ROC curves computed using this scheme as shown in Fig. 10 is highly densely packed and includes every operating point.

TABLE 1
Summary of the English and Arabic Evaluation Data Sets

	Tobacco-800	Maryland Arabic
Document Types	Printed/handwritten	Mostly handwritten
Total Pages	1290	169
Resolution (in DPI)	150–300	200
Labeled Signatures	900	149
Number of Signers	66	21

The multiscale saliency approach gives the best overall detection performance on both English and Arabic data sets. Using document context, our multiscale signature detector achieves 92.8 and 86.6 percent detection rates for the Tobacco-800 and Maryland Arabic data sets at 0.3 false positives per image (FPPI). Exploring context is shown to be much more effective on machine-printed documents as geometric relationships among text lines are uniform. Take the operating point of 0.1 FPPI as an example. Using context alone gives an average increase of 13.6 percent in detection accuracy on Tobacco-800 compared to only 5.1 percent on the Maryland Arabic database. The improvements by using multiscale are 5.8 and 4.0 percent on the Tobacco-800 and Maryland Arabic data sets, respectively. This demonstrates the advantage of multiscale approach on data sets that capture more diversity, such as the Tobacco-800 database.

Second, we test how discriminative our saliency measure is in capturing the dynamic curvature in signatures. The handwritten Maryland Arabic data set serves better for this purpose, because local features become clearly less discriminative as evident from the poor performance of the Fisher classifier.

Figs. 11 and 12 show samples of detected signatures from Tobacco-800 and Maryland Arabic data sets, together with their saliency maps. We show the top three most salient parts in red, green, and blue, respectively. In our experiment, a cursive structure is normally more than an order of magnitude more salient than printed text of the same dimensions. However, we did find rare instances of printed text among false positives, as shown in Fig. 13a, with comparable saliencies to signatures because of their highly cursive font styles. A limitation of our proposed method is that the detected and segmented signature may contain a few touching printed characters.

For better interpretation of the overall detection performance, we summarize key evaluation results. On Tobacco-800, 848 signatures out of the 900 labeled signatures are detected by the multiscale saliency approach using document

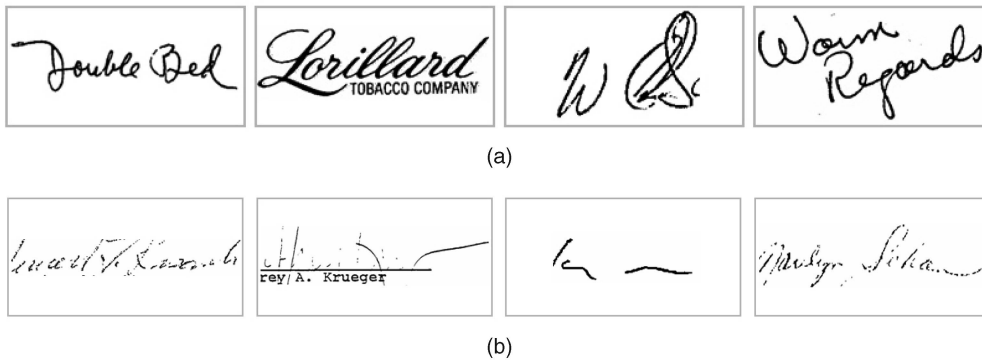


Fig. 13. Examples of (a) false alarms and (b) missed signatures from the Tobacco-800 database.

local maxima in gradient magnitude. A ridge detector can give more compact segmentation output since it produces only one thin curve in response. However, a ridge detector [60] performs much worse in signature detection in our experiments. This is because the Canny edge detector provides good localization that guarantees accurate estimation of local gradient directions and is more robust under structural variations.

Some examples of false positives from the Tobacco-800 set are shown in Fig. 13a, which include cases of handwriting. The classification between signature and handwriting is sometimes not well posed by considering shape alone. Highly cursive handwritten words may not have any obvious visual differences from signatures, as shown in Fig. 13a. Using geometric information cannot effectively resolve such intricacies in real documents because they are primarily annotations created in an ad hoc manner. Semantics and content layer information are required to solve the ambiguity in this case.

Fig. 13b shows examples of false negatives in detection. These missed signatures are severely broken and a step edge operator like Canny could not detect contours, even at a coarse scale. As shown in most signatures, however, using multiple scales for detection partially overcomes the limitations of a connected-components-based approach by relaxing the requirement that the contour fragment be well connected at a particular scale. This improvement is more clearly observed on the Tobacco-800 data set, which contains a considerable number of highly degraded images at low resolution.

6.4 Signature Matching for Document Image Retrieval

6.4.1 Shape Representation

We compare shape representations computed using different segmentation strategies in the context of document image retrieval. In particular, we consider skeleton and contour, which are widely used mid-level features in computer vision and can be extracted relatively robustly.

For comparison, we developed a baseline signature extraction approach by removing machine-printed text and noise from labeled signature regions in the groundtruth using a trained Fisher classifier [61]. To improve classification, the baseline approach models the local contexts among printed text using Markov Random Field (MRF). We implemented two classical algorithms—one by Dyer and Rosenfeld [62] and the other by Zhang and Suen [63], which compute skeletons from the signature layer extracted by the baseline approach. Fig. 14 illustrates the layer subtraction and skeleton extraction in the baseline approach as compared to the salient contours of detected and segmented signatures from documents by our approach.

In this experiment, we sample 200 points along the extracted skeleton and salient contour representations of each signature. We use the faster shape context matching algorithm to solve for correspondences between points on the two shapes and compute all the four shape distances using D_{sc} , D_{be} , D_{as} , and D_{re} . To remove any bias, the query signature is removed from the test set in that query for all retrieval experiments.

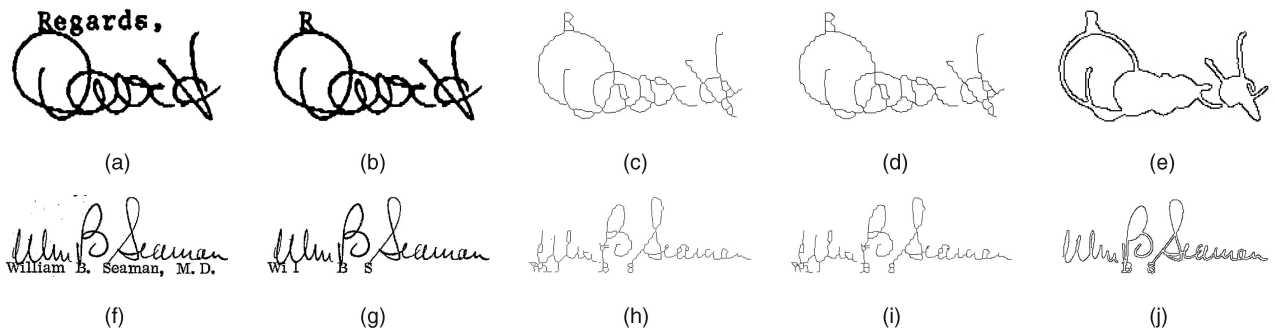


Fig. 14. Skeleton and contour representations computed from signatures. The first column represents labeled signature regions in the groundtruth. The second column represents signature layers extracted from labeled signature regions by the baseline approach [61]. The third and fourth columns are skeletons computed by Dyer and Rosenfeld [62] and Zhang and Suen [63], respectively. The last column represents salient contours of actual detected and segmented signatures from documents by our approach.

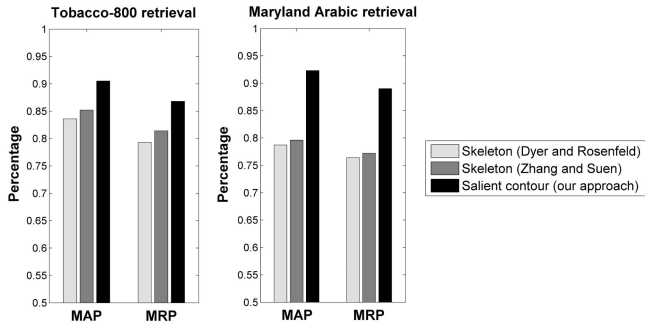


Fig. 15. Document image retrieval performance of different shape representations on the two data sets measured by MAP and mean R-precision (MRP).

Document image retrieval performance using different shape representations is shown in Fig. 15. Salient contours computed by our detection and segmentation approach outperform the skeletons that are directly extracted from labeled signature regions on both data sets. As illustrated in the third and fourth columns in Fig. 14, thinning algorithms are sensitive to structural variations among neighboring strokes and noise. In contrast, salient contours provide a globally consistent representation by weighting more on structurally important shape features. This advantage in retrieval performance is more evident on the Maryland Arabic data set in which signatures and background handwriting are closely spaced.

6.4.2 Shape Matching Algorithms

We developed signature matching approaches using two nonrigid shape matching algorithms—shape contexts and local neighborhood graph, and evaluate their retrieval performances on salient contours. We use all four measures of dissimilarity D_{sc} , D_{be} , D_{as} , and D_{re} in this experiment. The weights of different shape distances are optimized by linear discriminant analysis using randomly selected subset of signature samples as training data. Fig. 16 shows retrieval performances measured in MAP for both methods as the size of training set varies. A special case in Fig. 16 is when no training data are used. In this case, we simply normalize each shape distance by the standard deviation

TABLE 2
Retrieval Performance Using Different Measure of Shape Dissimilarity on the Tobacco-800 Database

Measure of Dissimilarity	MAP	MRP
D_{sc}	66.9%	62.8%
D_{as}	61.3%	57.0%
D_{be}	59.8%	55.6%
D_{re}	52.5%	48.3%
$D_{sc} + D_{be}$	78.7%	74.3%
$D_{sc} + D_{as} + D_{be} + D_{re}$	90.5%	86.8%

computed from all instances in that query, thus effectively weighting every shape distance equally.

A significant increase in overall retrieval performance is observed using only a fairly small amount of training data. Both shape matching methods are effective with no significant difference. In addition, the performances of both methods measured in MAP only deviate less than 2.55 and 1.83 percent, respectively, when different training sets are randomly selected. These demonstrate the generalization performance of representing signatures by nonrigid shapes and counteracting large variations among unconstrained handwriting through geometrically invariant matching.

6.4.3 Measures of Shape Dissimilarity

Table 2 summarizes the retrieval performance using different measures of shape dissimilarity on the larger Tobacco-800 database. The results are based on the shape context matching algorithm as it demonstrates smaller performance deviation in previous experiment. We randomly select 20 percent of signature instances as training data and use the rest for test.

The most powerful single measure of dissimilarity for signature retrieval is the shape context distance (D_{sc}), followed by the affine transformation-based measure (D_{as}), the TPS bending energy (D_{be}), and the registration residual error (D_{re}). By incorporating rich global shape

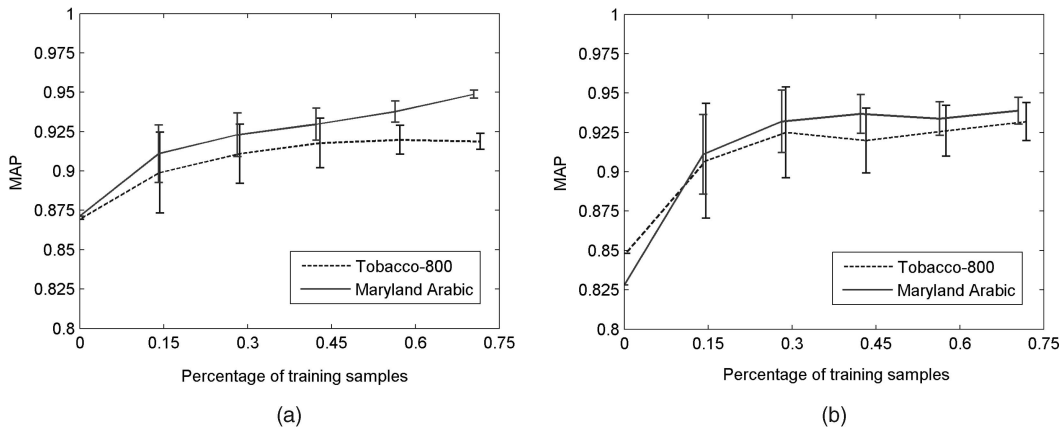


Fig. 16. Document image retrieval using single signature instance as (a) query using shape contexts [20] and (b) local neighborhood graph [21]. The weights for different shape distances computed by the four measures of dissimilarity can be optimized by LDA using a small amount of training data.

TABLE 3
Retrieval Performance Using Multiple Signature Instances
on the Tobacco-800 Database

Measure of Dissimilarity	MAP	MRP
One	90.5%	86.8%
Two	92.6%	88.2%
Three	93.2%	89.5%

information, shape contexts are discriminative even under large variations. Moreover, the experiment shows that measures based on transformations (affine for linear and TPS for nonlinear transformation) are also effective. The two proposed measures of shape dissimilarity, D_{sc} and D_{ber} , improve the retrieval performance considerably; increasing MAP improves it from 78.7 to 90.5 percent. This demonstrates that we can significantly improve the retrieval quality by combining effective complementary measures of shape dissimilarity through limited supervised training.

6.4.4 Multiple Instances as Query

Table 3 summarizes the retrieval performances using multiple signature instances as an equivalent class in each query on the Tobacco-800 database. The queries consist of all the combinations of multiple signature instances from the same person, giving even larger query sets. In each query, we generate a single ranked list of retrieved document images using the final shape distance between each equivalent class of query signatures and each searched instance defined in (17). As shown in Table 3, using multiple instances steadily improves the performance in terms of both MAP and MRP. The best results on Tobacco-800 are 93.2 percent MAP and 89.5 percent MRP when three instances are used for each query.

6.5 Offline Signature Verification

We quantitatively compare our signature matching approach with several state-of-the-art offline signature verification approaches [5], [9], [13] on the Sabourin offline signature database [9], which has been used in [5], [9], [64], [65]. This database of 800 signature images contains genuine and random forged signatures of 20 classes, with 40 signatures per class. Historically, results on the Sabourin signature database are reported in terms of false rejection rate (FRR) of genuine signatures and false acceptance rate (FAR) of forged signatures, with parameters trained to minimize total errors. We follow the same evaluation protocol as in [5], [9], where the first 20 signatures in each class are used for training and the rest 20 signatures for test.

Our approach obtained the state-of-the-art performance on offline signature verification on the Sabourin database without any explicit assumption on local features, such as granulometric size distributions in [9] and sets of stroke-level features in [5]. The 0.50 percent FRR and 0.21 percent FAR are the best published offline signature verification results on this database using the standard experimental protocol. As shown in Fig. 17, our approach effectively separates between genuine signatures and forgeries across all 20 signature classes. The performance of Sabourin et al. [9] in Table 4 is

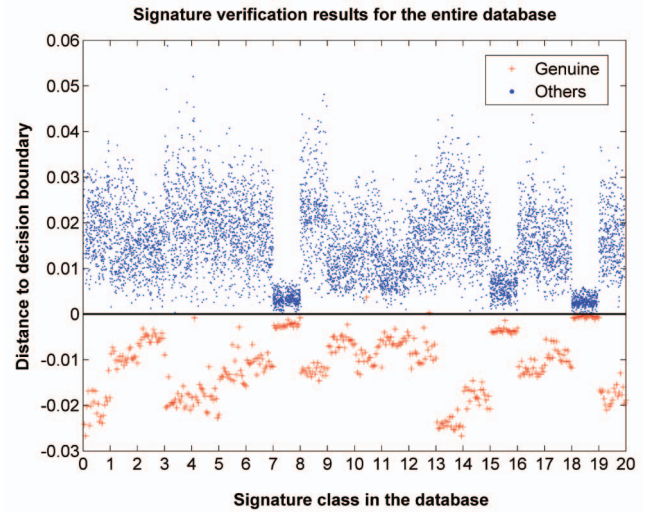


Fig. 17. Our approach robustly discriminates genuine signatures (red cross) from forgeries (blue dots) across all signature classes in the Sabourin signature verification database [9].

achieved by combining four classifiers using multiple classification schemes. The best single classifier reported in [9] gives 1.55 percent FRR and 0.15 percent FAR at 25 iterations. This experiment also demonstrates the big difference in performance between our approach and the approach of the authors of [13], which employs an improved dynamic time warping algorithm on 1D projection profile. Projection profile feature provides limited shape discriminability and is shown to be very sensitive to changes in the baseline direction. Heuristic alignment, such as by finding the orientation that gives the minimum length horizontal projection [13], is not robust under large structural variations in offline signatures. Treating signature in the unconstrained setting of a 2D deformable shape, our approach is more robust to large intraclass variations and gives a better generalization performance.

7 CONCLUSION

In this paper, we proposed a novel signature detection and segmentation approach based on the view that object detection can be a process that aims to capture the characteristic structural saliency of the object across image scales. This is different from the common object detection framework that focuses on sets of local properties. The results on signature detection using multilanguage document image

TABLE 4
Signature Verification Results Using
the Sabourin Offline Signature Database [9]

Approach	FRR	FAR
Sabourin <i>et al.</i> [9]	1.64%	0.10%
Guo <i>et al.</i> [5]	1.66%	1.10%
Shanker and Rajagopalan [13]	8.75%	3.26%
Our approach	0.50%	0.21%

collections show that our approach is effective in the presence of large variations and clustered backgrounds. One advantage of using the multiscale saliency approach for joint detection and segmentation is that it provides a general framework for which detection and segmentation degrade gracefully as the problem becomes more challenging. In addition, detected and segmented outputs are both structurally and perceptually meaningful for matching and recognition.

To robustly handle large structural variations, we treated the signature in the general setting of a nonrigid shape and demonstrated document image retrieval using state-of-the-art shape representations, measures of shape dissimilarity, shape matching algorithms, and by using multiple instances as query. We quantitatively evaluated these techniques in challenging retrieval tests, each composed of a large number of classes but relatively small number of signature instances per class. We also demonstrated our matching techniques in offline signature verification and achieved very competitive results. Extensive experiments demonstrated the excellent performance of our approaches.

APPENDIX

PROOF OF THEOREM 1

Theorem 1. *Given two intersecting lines $t_1(x, y)$ and $t_2(x, y)$ in the forms $p_1(x - x_1) + q_1(y - y_1) = 0$ and $p_2(x - x_2) + q_2(y - y_2) = 0$, respectively. Let $L(x, y)$ denote the straight line that passes through points (x_1, y_1) and (x_2, y_2) . Then, the quadratic equation $C(x, y) \equiv l^2(x, y) - \lambda t_1(x, y)t_2(x, y) = 0$ represents a conic section and the range of λ for which $C(x, y)$ is an ellipse is $0 < \lambda < \lambda_0$, where*

$$\lambda_0 = \frac{4[p_1(x_2 - x_1) + q_1(y_2 - y_1)]}{(p_1q_2 - p_2q_1)} \times \frac{[p_2(x_1 - x_2) + q_2(y_1 - y_2)]}{(p_1q_2 - p_2q_1)}.$$

Proof. The canonical form of a quadratic equation $C(x, y)$ in two variables x and y is

$$C(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0, \quad (\text{A.1})$$

where a, b, c, f, g, h are linear functions in λ and the parameter set $(x_1, y_1), (x_2, y_2), (p_1, q_1), (p_2, q_2)$. We can rewrite (A.1) as

$$\mathbf{x}^T \mathbf{A} \mathbf{x} + 2\mathbf{f}^T \mathbf{x} + c = 0, \quad (\text{A.2})$$

where $\mathbf{x} = (x, y)^T$, $\mathbf{f} = (g, f)^T$, and the 2×2 matrix $\mathbf{A}$ is

$$\mathbf{A} = \begin{pmatrix} a & h \\ h & b \end{pmatrix}. \quad (\text{A.3})$$

When $C(x, y)$ is an ellipse with center $\mathbf{x}_0$ at $(x_0, y_0)^T$, (A.2) can be written as

$$(\mathbf{x} - \mathbf{x}_0)^T \mathbf{A} (\mathbf{x} - \mathbf{x}_0) = d, \quad (\text{A.4})$$

i.e.,

$$\mathbf{x}^T \mathbf{A} \mathbf{x} - 2\mathbf{x}_0^T \mathbf{A} \mathbf{x} + \mathbf{x}_0^T \mathbf{A} \mathbf{x}_0 - d = 0. \quad (\text{A.5})$$

Comparing (A.2) with (A.8), we have that

$$\mathbf{A} \mathbf{x}_0 = -\mathbf{f},$$

giving

$$\mathbf{x}_0 = -\mathbf{A}^{-1} \mathbf{f} \quad (\text{A.6})$$

and

$$d = \mathbf{f}^T \mathbf{A}^{-1} \mathbf{f} - c. \quad (\text{A.7})$$

Equation (A.5) represents a parabola if $\mathbf{A}$ is singular since then the center $\mathbf{x}_0$ is at infinity. It represents an ellipse if $\mathbf{A}$ is positive definite. When $\lambda = 0$, the matrix $\mathbf{A}$ is singular. As λ increases, $\mathbf{A}$ becomes positive definite first and then becomes singular again at a certain positive value λ_0 . For λ larger than λ_0 , matrix $\mathbf{A}$ is indefinite and (A.5) becomes a hyperbola. Thus, the range of λ_0 for which (A.5) is an ellipse is the interval $0 < \lambda < \lambda_0$. We can find λ_0 by solving

$$\det(\mathbf{A}) = \begin{vmatrix} a & h \\ h & b \end{vmatrix} = 0, \quad (\text{A.8})$$

where

$$a(\lambda) = (y_1 - y_2)^2 - \lambda p_1 p_2, \quad (\text{A.9})$$

$$b(\lambda) = (x_2 - x_1)^2 - \lambda q_1 q_2, \quad (\text{A.10})$$

$$h(\lambda) = (y_1 - y_2)(x_2 - x_1) - \frac{1}{2} \lambda (p_1 q_2 + p_2 q_1). \quad (\text{A.11})$$

One root of (A.8) is zero and the other root is strictly positive. The positive root is the required λ_0 and given by

$$\lambda_0 = \frac{4[p_1(x_2 - x_1) + q_1(y_2 - y_1)]}{(p_1q_2 - p_2q_1)} \times \frac{[p_2(x_1 - x_2) + q_2(y_1 - y_2)]}{(p_1q_2 - p_2q_1)}. \quad (\text{A.12})$$

□

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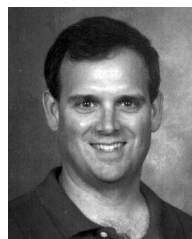
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